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of values of all four quantum numbers." It means that, each electron must have unique values for the four quantum numbers (n, *l*, m and s).

For the lone electron present in hydrogen atom, the four quantum numbers are: n = 1; l = 0; m = 0 and s = +½. For the two electrons present in helium, one electron has the quantum numbers same as the electron of hydrogen atom, n = 1, *l* = 0, m = 0 and s = +½. For other electron, the fourth quantum number is different i.e., n = 1, *l* = 0, m = 0 and s = –½.

As we know that the spin quantum number can have only two values +½ and –½, only two electrons can be accommodated in a given orbital in accordance with Pauli exclusion principle. Let us understand this by writing all the four quantum numbers for the eight electron in L shell.

Table 2.3 Quantum numbers of electrons in L shell

Electron	n	l	m	s
1st	2	0	0	+½
2nd	2	0	0	-½
3rd	2	1	-1	+½
4th	2	1	-1	-½
5th	2	1	0	+½
6th	2	1	0	-½
7th	2	1	+1	+ ½
8th	2	1	+1	-½

2.6.3 Hund's rule of maximum multiplicity

The Aufbau principle describes how the electrons are filled in various orbitals. But the rule does not deal with the filling of electrons in the degenerate orbitals (i.e. orbitals having same energy) such as p_x, p_y and p_z. In what order these orbitals to be filled? The answer is provided by the Hund's rule of maximum multiplicity. It states that electron pairing in the degenerate orbitals does not take place until all the available orbitals contains one electron each.

We know that there are three p orbitals, five d orbitals and seven f orbitals. According to this rule, pairing of electrons in these orbitals starts only when the 4th, 6th and 8th electron enters the p, d and f orbitals respectively.

For example, consider the carbon atom which has six electrons. According to Aufbau principle, the electronic configuration is $1s^2$, $2s^2$, $2p^2$

It can be represented as below,



In this case, in order to minimise the electron-electron repulsion, the sixth electron enters the unoccupied $2p_y$ orbital as per Hunds rule. i.e. it does not get paired with the fifth electron already present in the $2P_x$ orbital.

Evaluate Yourself



6. How many unpaired electrons are present in the ground state of Fe^{3+} (z=26), Mn^{2+} (z=25) and argon (z=18)?